



Riphah International University Sahiwal

Riphah College of Computing and Innovation

Instructor: Dr. M. Asad Iqbal | **Course Title:** Linear Algebra (MT-2303)

Instructions

1. Don't use any AI source to solve this assignment.
2. Don't copy from any of your colleague's assignment.
3. Use A4 page pages and submit handwritten assignment.
4. Strictly follow the submission deadline.

Assignment # 1

Case: A university uses two simple indicators to decide whether an applicant should be admitted to a BS Artificial Intelligence program:

1. x = Entry test performance
 - 0 = Low
 - 1 = Average
 - 2 = High
2. y = Academic performance
 - 0 = Low
 - 1 = Average
 - 2 = High

The admission system classifies each student as Admit or Not Admit.

Students	Entry-Test (x)	Academic (y)	Admission
A	2	2	admit
B	2	1	admit
C	1	2	admit
D	1	1	not admit
E	0	2	not admit
F	2	0	not admit
G	0	1	not admit
H	1	0	not admit
I	2	2	admit

Task 1: Construct a classifier with suitable threshold (use method other than the vector dot product).

Task 2: Choose a suitable model, then use vectors dot product to predict the outputs.

Task 3: Construct a neural network using above data points and explain its features using python.